

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – X
PAPER –1
TEST DATE: 21-08-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 – 04, 20 – 23, 39 – 42): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
Section – A (05 – 10, 24 – 29, 43 – 48): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

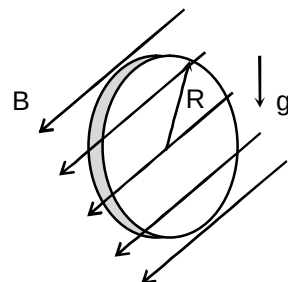
Physics

PART – I

Section – A (Maximum Marks: 12)

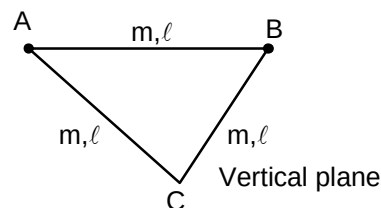
This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. A thin conducting circular disc of mass m , thickness d and radius R ($R \gg d$) is released in a uniform horizontal magnetic field of induction $\vec{B}$. The magnetic induction vector $\vec{B}$ is in the plane of the disc as shown in the figure. The acceleration of the disc is



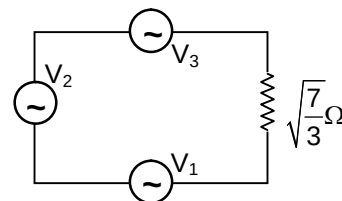
- (A) $\frac{mg}{(m + 2\epsilon_0\pi R^2 B^2 d)}$
 (B) $\frac{mg}{(2m + \epsilon_0\pi R^2 B^2 d)}$
 (C) $\frac{2mg}{(m + \epsilon_0\pi R^2 B^2 d)}$
 (D) $\frac{mg}{(m + \epsilon_0\pi R^2 B^2 d)}$

2. Three rigid rods each of mass m and length ℓ are joined to form an equilateral triangle. The two corners A and B are pivoted in the same horizontal level and this assembly is kept in a vertical plane. The hinge reaction at A immediately after nail B is removed.



- (A) $\sqrt{3}mg$
 (B) $2\sqrt{3}mg$
 (C) $3\sqrt{3}mg$
 (D) $4\sqrt{3}mg$

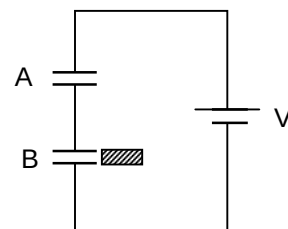
3. Three alternating voltage sources $V_1 = 3 \sin \omega t$ volt, $V_2 = 5 \sin(\omega t + \phi_1)$ volt and $V_3 = 5 \sin(\omega t - \phi_2)$ volt connected across a resistance $R = \sqrt{\frac{7}{3}} \Omega$ as shown in the figure. (Where ϕ_1 and



ϕ_2 correspond) to 30° and 127° respectively). The peak current through the resistor is

- (A) 1A
 (B) 2A
 (C) 3A
 (D) 4A

4. Two identical capacitors A and B shown in the given circuit are joined in series with a battery. If a dielectric slab of dielectric constant K is inserted between the plates of capacitor B and battery remains connected, then the energy stored in the capacitor A will

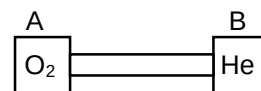


- (A) decrease
 (B) increase
 (C) remain the same
 (D) initially increase and then decrease to initial value

Section – A (Maximum Marks: 24)

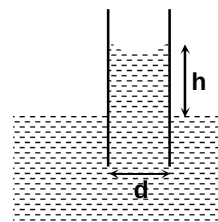
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. Two containers A and B are connected by a conducting solid cylindrical rod of length $\frac{242}{7}$ cm and radius $\sqrt{8.3}$ cm. Thermal conductivity of the rod is 693 W/m-K and specific heat capacity of the rod is negligible. The container A contains two moles of oxygen gas and container B contains four moles of helium gas. At time $t = 0$ temperature difference of the containers is 50°C , after what time temperature difference between them will be 25°C . Transfer of heat takes place through the rod only. Neglect radiation loss. (Take $R = 8.3$ J/mole-K, $\pi = \frac{22}{7}$ and $\ln 2 = 0.693$)



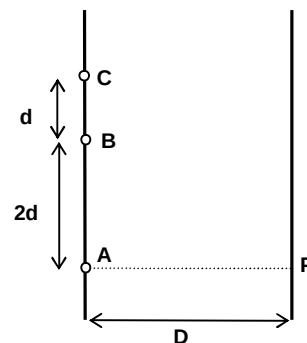
- (A) 3 sec
(B) 5 sec
(C) 7 sec
(D) 9 sec
6. A uniform electric field exists in xy plane. The potential at points A(2m,2m), B(-2m,2m) and C(2m,4m) are 4V, 16V, and 12V respectively. The components of electric field in x and y directions are
- (A) $E_x = 3$ V/m
(B) $E_x = 4$ V/m
(C) $E_y = 4$ V/m
(D) $E_y = -4$ V/m

7. Two vertical parallel rectangular glass plates are partially submerged in water. The separation between the plates is $d = 0.1$ mm and the width of each plate is $\ell = 10$ cm. Assuming that the water between the plates does not reach the upper edges of the plates and the contact angle is $\theta = 0$. The surface tension of water is $T = 7 \times 10^{-2}$ N/m and the density of water is $\rho = 10^3$ kg/m³. Then choose the correct option(s). ($g = 10$ m/s²)



- (A) The water rises upto height $h = 14$ cm between the plates.
(B) The water rises upto height $h = 28$ cm between the plates.
(C) The force of mutual attraction between the plates is 4.9 N.
(D) The force of mutual attraction between the plates is 9.8 N.

8. In the figure A, B and C are three slits each of them individually producing the same intensity I_0 at P when they are illuminated by parallel beam of light of wavelength λ . It is given that $BP - AP = \frac{\lambda}{2}$. Also given that $d \ll D$, then wavelength λ and the resulting intensity I at P will be ($\lambda \ll d$).



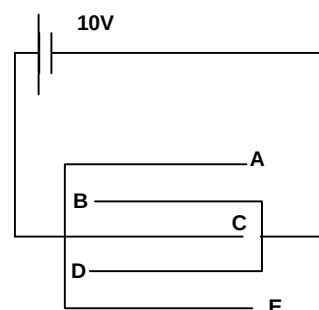
- (A) $\lambda = \frac{2d^2}{D}$
(B) $\lambda = \frac{4d^2}{D}$
(C) $I = 2I_0$
(D) $I = I_0$

9. Which of the following products in Bohr's atom are independent of principal quantum number n ? ($v \rightarrow$ velocity, $E \rightarrow$ energy, $r \rightarrow$ radius)
- (A) vn
 (B) Er
 (C) En
 (D) vr
10. In a hypothetical atom, potential energy between electron and nucleus is given by $U = k\ell nr$. Where r is orbital radius and k is a constant. The expression for orbital radius and total energy for such an atom according to Bohr's model are
- (A) $r = \frac{nh}{2\pi\sqrt{mk}}$
 (B) $r = \frac{nk}{2\pi\sqrt{mk}}$
 (C) $E = \left[\frac{1}{2} + \ell n \left(\frac{nh}{2\pi\sqrt{mk}} \right) \right] k$
 (D) $E = k\ell n \left(\frac{nh}{2\pi\sqrt{nk}} \right)$

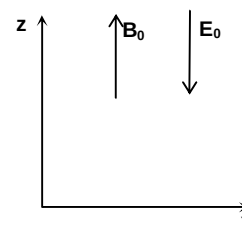
Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

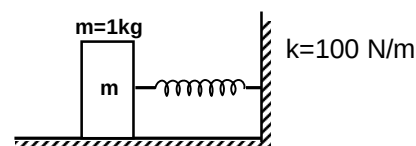
11. Five identical capacitor plates are arranged such that they make four capacitors each of $2\mu\text{F}$. The plates are connected to a source of emf 10 V. Find the charge on the plate C in μC .



12. In a certain region, uniform electric field $E = -E_0\hat{k}$ and magnetic field $B = B_0\hat{k}$ are present. At time $t = 0$, a particle of mass m and charge q is given a velocity $v = v_0\hat{j} + v_0\hat{k}$. Find the time (in sec) when the speed of the particle becomes minimum. (Given $mv_0 = 4qE_0$)



13. A spring mass system is kept on a rough horizontal surface having co-efficient of friction $\mu = 0.4$. One end of the spring is fixed to the vertical wall as shown in the figure. Initially the spring is stretched by 27 cm and then it is released. The total distance covered by the block before it comes to permanent rest is x cm. Find x . (Take $g = 10 \text{ m/s}^2$)

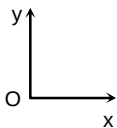


Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15**Question Stem**

The space has electromagnetic field which varies with time whose variation is given as:

$$\vec{B} = \begin{cases} -B_0 \hat{k} & \text{if } 0 \leq t \leq \frac{\pi m}{qB_0} \\ -B_0 \hat{j} & \text{if } \frac{\pi m}{qB_0} \leq t \leq \frac{2\pi m}{qB_0} \\ 0 & \text{if } \frac{2\pi m}{qB_0} \leq t < \infty \end{cases} \text{ and } \vec{E} = \begin{cases} 0 & \text{if } 0 \leq t \leq \frac{2\pi m}{qB_0} \\ -E_0 \hat{k} & \text{if } \frac{2\pi m}{qB_0} \leq t < \infty \end{cases}$$


A charge particle having mass m and positive charge q is given a velocity $v_0 \hat{i}$ at origin at $t = 0$. The coordinates of the point on xy plane where it again passes through the xy plane for the first time are

$$\left(\frac{Kmv_0}{q} \sqrt{\frac{v_0}{E_0 B_0}}, \frac{2mv_0}{NqB_0}, 0 \right).$$

14. The value of K is

15. The value of N is

Question Stem for Question Nos. 16 and 17**Question Stem**

Ring, solid cylinder and solid sphere are placed on the top of a rough inclined plane of inclination θ and all of them are released from rest at same instant from same point. Only solid sphere can just roll without

slipping. The ratio of their time to reach the ground $\left(\frac{t_{\text{ring}}}{t_{\text{sphere}}} \right)$ is X and the ratio of their time to reach the

ground $\left(\frac{t_{\text{cylinder}}}{t_{\text{sphere}}} \right)$ is Y .

16. The value of X is.....

17. The value of Y is.....

Question Stem for Question Nos. 18 and 19

Question Stem

A river has width d . A fisherman in a boat crosses the river twice. During the first crossing, his goal is to minimize the crossing time. During the second crossing, his goal is to minimize the distance that the boat is carried downstream. In the first case, the crossing time is t while in the second case, the crossing time is $3t$. The speed of fisherman is $\left(\frac{1}{X}\right)\frac{d}{t}$ and the speed of river is $\left(\frac{Y}{2\sqrt{2}}\right)\frac{d}{t}$.

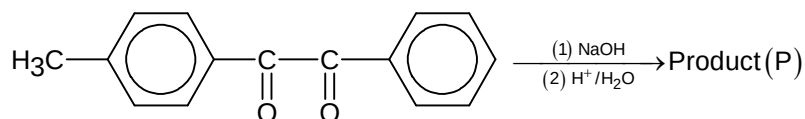
18. The value of X is

19. The value of Y is

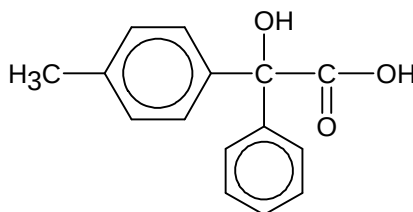
Chemistry**PART – II****Section – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

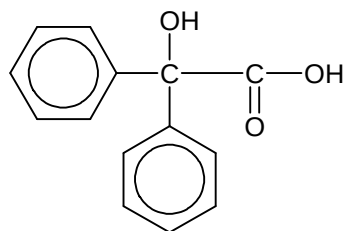
20. The major product of the reaction is



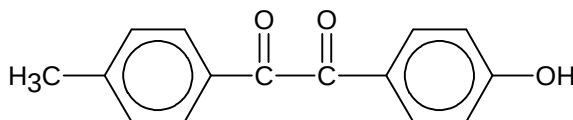
(A)



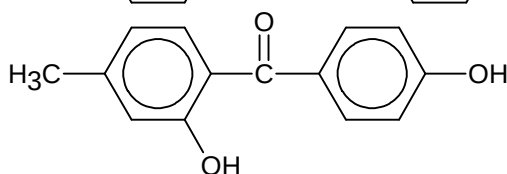
(B)



(C)



(D)



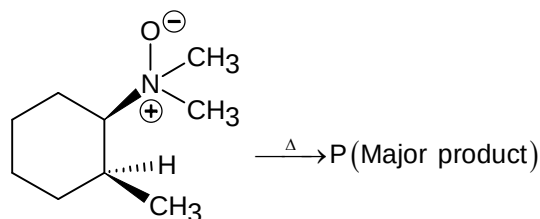
21. Beryl ($\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$) is a silicate. The number of oxygen atom of structural unit of SiO_4^{4-} shared with neighbouring unit in Beryl are:

(A) 1
(B) 2
(C) 3
(D) 4

22. Which of the following statements is incorrect?

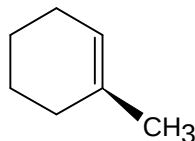
(A) Glycerol (1 mole) on reaction with periodic acid (2 moles) yields formaldehyde and formic acid
(B) Glycerol on dehydration in presence of conc. H_2SO_4 yields unpleasant smelling compound
(C) Oxidation of glycerol with conc. HNO_3 gives mainly glyceric acid
(D) Glycerol on heating with large excess of HI produces allyl iodide

23.

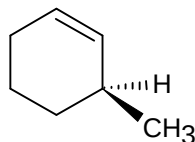


P is

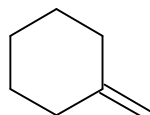
(A)



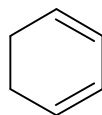
(B)



(C)



(D)

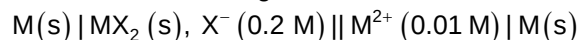


Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

24.

Consider the following cell:



The EMF of the cell at 298 K is 0.236 V

 ($\text{MX}_2(\text{s})$ being the sparingly soluble salt):

Pick out the correct options of the following:

- (A) The solubility product of MX_2 is $2.0 \times 10^{-11} \text{ M}^3$
- (B) The solubility of MX_2 in 0.2 M X^- is $1.0 \times 10^{-10} \text{ M}$
- (C) The solubility product of MX_2 is $4.0 \times 10^{-12} \text{ M}^3$
- (D) The solubility of MX_2 in pure water is $1.0 \times 10^{-4} \text{ M}$

25.

The correct statement(s) regarding cellulose is/are

- (A) It is polymer of D-glucose
- (B) It has $\beta - 1,4$ – glycosidic linkage
- (C) It has general formula $(\text{C}_6\text{H}_{10}\text{O}_5)_n$
- (D) It is used for making rayon fibre

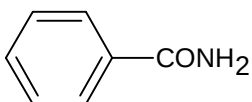
26.

Which of the following pairs consists of ore of same metal and both are oxides?

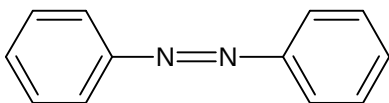
- (A) Haematite, Magnetite
- (B) Anglesite, Cerrusite
- (C) Bauxite, Corundum
- (D) Cuprite, Cassiterite

27. Kjeldahl's method cannot be used for estimation of nitrogen present in

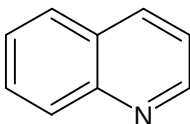
(A)



(B)



(C)



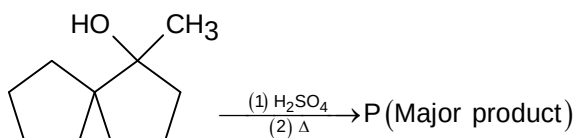
(D)



28. In container 'A', an ice cube of pure D_2O is to be placed in pure liquid H_2O . In container 'B', an ice cube of pure H_2O is to be placed in pure liquid D_2O . Which of the following is/are correct statements?

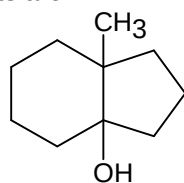
- (A) Ice cube will float in both containers A and B.
 (B) Ice cube will sink in container A.
 (C) Ice cube will sink in both containers A and B.
 (D) Ice cube will float over in container B.

29.

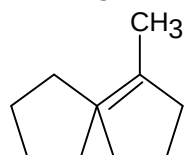


Products are

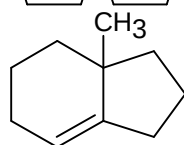
(A)



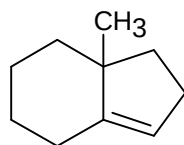
(B)



(C)



(D)



Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

30. What is the total number of correct statements in the following?
 (A) Antiseptics are applied to living tissues.
 (B) Aspirin is non-narcotic analgesic.
 (C) Disinfectants are applied to inanimate.
 (D) Aspartame is an artificial sweetener.
 (E) Aspirin is acetyl salicylic acid.
 (F) Insulin is a co-enzyme.
 (G) Aluminium hydroxide is an antacid.
 (H) D.D.T. is non-biodegradable pollutant.
31. If P, Q, R, S is the total number of unpaired electrons in t_{2g} or t_2 set of orbitals in $K_3[Fe(CN)_6]$, $[FeCl_4]^{2-}$, $[Fe(H_2O)_6]Cl_3$, $[FeF_6]^{4-}$ respectively, then what is the value of ' $P + Q + R + S$ '?
32. If closed vessel contains 5 g of ice at $-10^\circ C$, then how much heat in kJ must be added to convert it into steam at $205.56^\circ C$
 [Given: Specific heat capacity (ice) = $2.00 \text{ J/g}^\circ C$
 Specific heat capacity of water = $4.18 \text{ J/g}^\circ C$
 Specific heat capacity of steam = $1.8 \text{ J/g}^\circ C$
 Enthalpy of fusion of water = 333 J/g
 Enthalpy of vaporization of water = 2439 J/g

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem

One mole each of ammonium di-chromate, ammonium nitrate and mercuric nitrate are heated strongly and the gases evolved are collected. If 'X' is the total volume of diatomic gas/gases (in litres at S.T.P.) and 'Y' is the total volume of triatomic gas/gases (in litres at S.T.P.), (molar volume of a gas at STP is 22.4 L), then

33. The value of 'X' is _____
34. The value of 'Y' is _____

Question Stem for Question Nos. 35 and 36**Question Stem**

A solution of 1.660 g of KI in 100 g of water freezes at 'X' Kelvin. The freezing point of this solution saturated with solid HgI_2 (Sparingly soluble in water) is 'Y' Kelvin. Considering KI and any species formed to be completely ionized, answer the following.

[Given: $K_{f(\text{water})} = 1.86 \text{ K kg mol}^{-1}$, freezing point of water = 273.15 K, Atomic mass of K = 39, I = 127]

35. What is the value of Y(in Kelvin)?
36. What is the value of Y – X (in Kelvin)?

Question Stem for Question Nos. 37 and 38**Question Stem**

A hydrogen-like species (atomic number Z) is in an excited state of quantum number n. An energy 3.40 eV is required to remove the electron completely from this excited atomic species. Also, this excited atom can make a transition to the II^{nd} excited state by successively emitted two photons of energies 4.25 eV and 5.95 eV respectively.

37. What is the value of 'Z'?
38. What is the ionization energy of first excited state of the atomic species?

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

39. Consider two curves $C_1 : y^2 = 4 \left[\sqrt{y} \right] x$ and $C_2 : x^2 = 4 \left[\sqrt{x} \right] y$ where $[.]$ denotes the greatest integer function, then the area of region enclosed by these two curves within the square formed by the lines $x = 1, y = 1, x = 4, y = 4$
- (A) $\frac{11}{9}$
 (B) $\frac{11}{7}$
 (C) $\frac{11}{5}$
 (D) $\frac{11}{3}$
40. A differentiable function g satisfy $\int_0^x (x-t+1)g(t)dt = x^4 + x^2 \quad \forall x \geq 0$, then $g(x)$ is
- (A) $12x^2 - 24x + 26 - 26e^{-x}$
 (B) $12x^2 + 24x + 26 - 26e^{-x}$
 (C) $12x^2 + 24x - 26 - 26e^{-x}$
 (D) $12x^2 - 24x - 26 - 26e^{-x}$
41. For a real number a , let $[a]$ denote the greatest integer less than or equal to a . Let R denote the region in the coordinate plane consisting of points (x, y) such that $[x]^2 + [y]^2 = 25$, the region R is completely contained in a disk of radius r (a disk is the union of a circle and its interior), the minimum values of r can be
- (A) $\frac{\sqrt{130}}{2}$
 (B) $\sqrt{\frac{130}{2}}$
 (C) $\frac{130}{2}$
 (D) $\frac{\sqrt{130}}{4}$
42. Maya lists all the positive divisors of 2010^2 . She then randomly selects two distinct divisors from this list. Then the probability that exactly one of the selected divisors is a perfect square is
- (A) $\frac{32}{81}$
 (B) $\frac{26}{81}$
 (C) $\frac{16}{81}$
 (D) $\frac{34}{81}$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

43. A pyramid with vertex at the point P, whose position vector is $4\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k}$ has a regular hexagonal base ABCDEF. Position vectors of a point A and B are $\hat{i}$ and $\hat{i} + 2\hat{j}$ respectively centre of the base has the position vector $\hat{i} + \hat{j} + \sqrt{3}\hat{k}$. Altitude drawn from P on the base meets the diagonal AD at G, then the possible position vectors of G will be, it is given that volume of the pyramid is $6\sqrt{3}$ cubic units
- (A) $\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k}$
 (B) $\hat{i} + 2\hat{j} + \sqrt{3}\hat{k}$
 (C) $\hat{i} - 2\hat{j} - 2\sqrt{3}\hat{k}$
 (D) $3\hat{i} - 2\hat{j} - 2\sqrt{3}\hat{k}$
44. Integral value of $x \in [0, 2\pi]$ which satisfy the inequality $2\cos x \leq \left| \sqrt{1+\sin 2x} - \sqrt{1-\sin 2x} \right| \leq \sqrt{2}$ is
- (A) 1
 (B) 2
 (C) 3
 (D) 4
45. If the quadratic equation $4^{\sec^2 A} x^2 + 2x + \left(B^2 - B + \frac{1}{2} \right) = 0$ have real roots, then the possible value of $\cos A + \cos^{-1} B$ is
- (A) $\frac{\pi}{3}$
 (B) $\frac{\pi}{3} + 1$
 (C) $\frac{\pi}{3} - 1$
 (D) $-\frac{\pi}{3}$
46. Let $\vec{r} = \vec{a} + \lambda\vec{\alpha}$ and $\vec{r} = \vec{b} + \mu\vec{\beta}$ are two lines through the point A and B with position vectors $\vec{a}$ and $\vec{b}$ respectively, then position vector of P and Q are where PQ is the shortest distance between the given lines
- (A) $\vec{P} = \vec{a} + \frac{((\vec{a} - \vec{b}) \cdot (\vec{\beta} \times (\vec{\beta} \times \vec{\alpha}))) \vec{\alpha}}{(\vec{\alpha} + \vec{\beta})^2}$
 (B) $\vec{P} = \vec{a} - \frac{((\vec{a} - \vec{b}) \times (\vec{\beta} \times (\vec{\beta} \times \vec{\alpha}))) \vec{\alpha}}{(\vec{\alpha} + \vec{\beta})^2}$
 (C) $\vec{Q} = \vec{b} + \frac{((\vec{a} - \vec{b}) \cdot (\vec{\alpha} \times (\vec{\beta} \times \vec{\alpha}))) \vec{\beta}}{(\vec{\alpha} + \vec{\beta})^2}$

$$(D) \quad \vec{Q} = \vec{b} - \frac{[(\vec{a} - \vec{b}) \cdot (\vec{a} \times (\vec{\beta} \times \vec{a}))] \vec{\beta}}{[(\vec{a} + \vec{\beta})]^2}$$

47. There are real numbers a, b, c and d such that -20 , is a root of $x^3 + ax + b$ and -21 is a root of $x^3 + cx^2 + d$. These two polynomials have a complex root $m + \sqrt{n}i$, where m, n are positive integers and $i = \sqrt{-1}$, then m and n can be
- (A) $m = 10$
 (B) $m = 320$
 (C) $n = 320$
 (D) $n = 10$

48. Let $f(x) = \begin{cases} (1-x)^a x(1-\cos 2\pi x) & \text{if } 0 < x \leq 1 \\ 0 & \text{if } x = 0 \end{cases}$, if Rolle's theorem is applicable to $f(x)$ for $x \in [0, 1]$, then a , can be
- (A) -2
 (B) -1
 (C) $\frac{1}{2}$
 (D) 1

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. Let $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ and $g(x) = f^{-1}(x)$, then $g'''(0)$ is
50. Let ABC be a triangle such that $AB = 7$, and let the angle bisector of $\angle BAC$ intersect line BC at D . If there exist point E and F on sides AC and BC , respectively such that lines AD and EF are parallel and divide triangle ABC into three parts of equal area, determine the number of possible integral values of BC
51. The value of $3p$, so that $\lim_{x \rightarrow \infty} x^p (\sqrt[3]{x+1} + \sqrt[3]{x-1} - 2 \cdot \sqrt[3]{x})$ is equal to some non-zero real number

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53

Question Stem

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f'(0) = 1$ and $f(x + 2y) = f(x) + f(2y) + e^{x+2y}(x + 2y) - x \cdot e^x - 2y e^{2y} + 4xy$ for all $x, y \in \mathbb{R}$, then

52. The value of $f(2) - 2e^2$ is
53. The value of $f'(3) - 4e^3$ is

Question Stem for Question Nos. 54 and 55**Question Stem**

Let C_1 be the curve $xy = 1$ and C_2 the reflection of C_1 in the line $y = 2x$ let the equation of C_1 be written in the form of $12x^2 + bxy + cy^2 + d = 0$

54. Then the value of b.c. is

55. The value of d is

Question Stem for Question Nos. 56 and 57**Question Stem**

A function f is defined for all real numbers and satisfy $f(2 + x) = f(2 - x)$ and $f(7 + x) = f(7 - x)$ for all real x

56. If $x = 0$ is a root of $f(x) = 0$, then the least number of roots of $f(x) = 0$ in the interval $[-1000, 1000]$ is

57. The period of $f(x)$ is